

An on-device AI exercise coach — private, offline, real-time.

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PROBLEM STATEMENT

Form-correct exercise needs real-time feedback — which normally means a personal trainer, a wearable, or a cloud vision API: expensive, and privacy-invasive (your camera streams to someone else's server). Home, rural and rehab users get neither safe form correction nor reliable rep counting. Our goal: trainer-grade coaching that runs **entirely on the phone**, with the camera feed never leaving the device.

TECH STACK + SOLUTION DEVELOPED

Stack: Android · Java · Camera2 · OpenCV · Qualcomm SNPE → Hexagon NPU (INT8 .dlc) · YOLO-NAS + HRNet (17 keypoints) · pure-Java analysis engine · on-device TextToSpeech · SharedPreferences + AlarmManager · no backend, no network, no INTERNET permission.

Solution: live skeleton tracking → auto-recognises 7 exercises (squat, push-up, curl, shoulder-press, sit-up, jumping-jack, plank) → counts reps + scores form 0–100 + live cues → an **offline voice coach** that speaks pattern-based corrections every few reps → offline profiles: personal bests, streaks, reminders.

SOLVING THE HARD PARTS

- **Recognition under flicker & noise** — a sticky, self-correcting lock + One-Euro keypoint smoothing + teleport/dropout rejection: reps survive jitter, and a wrong first guess corrects itself.
- **Counting that fits real bodies** — a two-threshold rep state-machine + adaptive per-user range calibration + peak/valley completion, so partial-range and no-lockout reps still count — and fast reps at low frame-rate are never dropped.
- **Camera-angle robustness** — a viewpoint-stable hip-drop signal counts foreshortened, front-on squats that a knee angle alone misses.
- **Zero misreads** — positive-evidence gates: a sit-up's trunk-fold can never be mistaken for a bicep curl or a shoulder press.
- **Real-time on a phone** — zero heap allocation per frame (pre-allocated ring buffers) so the GC never stutters mid-rep; INT8 on the NPU keeps it fast and battery-light, with a one-tap GPU/CPU fallback.
- **Proven, not hand-waved** — the entire engine is pure-Java and validated by a 128-assertion offline test harness that runs in milliseconds.

HOW IT WORKS — EVERY STAGE ON-DEVICE



SOCIAL RELEVANCE + POTENTIAL IMPACT

WHO IT REACHES

- **Access & equity** — trainer-grade form for the millions with no gym, coach or PT.
- **Privacy & dignity** — camera never leaves the phone — safe for women, minors, clinical & rehab use.
- **Rehab & ageing** — guided reps between physiotherapy visits; elderly home workouts.
- **Reach** — fully offline — tier-2/3, rural, low-connectivity; budget phone to flagship.

POTENTIAL IMPACT

- **Fewer injuries, real adherence** — live form turns risky reps into safe ones.
- **Remote physiotherapy** — track recovery at home; ease the load on clinics.
- **Privacy-first health AI** — a template: useful AI that collects zero data.
- **~Zero marginal cost** — no servers, no cloud bill, no data-centre energy; millions, on-device.